

Numerical Methods for Parameter Estimation in Non-linear Differential Algebraic Equations

Hans Georg Bock, Ekaterina Kostina, and Johannes P. Schlöder*

Interdisciplinary Center for Scientific Computing (IWR), Heidelberg University, Im Neuenheimer Feld 368, 69120 Heidelberg, Germany

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Nonlinear Differential Algebraic Equations (DAEs) are an important class of models for dynamic processes. To establish models that describe the process behavior in a quantitatively correct way, often parameters in the model have to be determined from observations or measurements of the process.

This paper reviews numerical methods for parameter estimation in DAEs. In particular the so-called boundary value problem approach and a versatile realisation, the multiple shooting method for parameter estimation, are discussed. Several applications are given to show the numerical performance and the wide applicability of the methods. A difficulty that occurs in practical applications is that the experiments performed to obtain measurements for parameter estimation are expensive, but nevertheless do not guarantee satisfactory parameter accuracy. The optimization of one or more dynamic experiments in order to maximize the accuracy of the results of a parameter estimation subject to cost and further technical inequality constraints leads to very complex non-standard optimal control problems. Newly developed methods for design of optimal experiments for nonlinear processes are briefly discussed.

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1 Introduction

Nonlinear differential equations play an important role in modelling dynamic processes in science, engineering or medicine not only in academia but of increasing importance also in industry and commerce. The establishment of such models involves joining theory and experimental or observational data, which includes the task of determining model parameters in such a way that the model reproduces the data best. Quantitatively correct models are in particular important, if the models are further used in design optimization or optimal control.

Efficient numerical methods for parameter estimation in a general class of DAE boundary value problems are subject of this paper. We discuss several features of parameter estimation problems and review methods for their solution. Emphasis is put on so-called boundary value problem methods with its versatile realization in multiple shooting.

* Corresponding author: e-mail: schloeder@iwr.uni-heidelberg.de, Phone: +49 6221 54 8239, Fax: +49 6221 54 5444

Nonlinear experimental design methods are presented that allow to optimally determine design and operation of experiments that deliver data with maximum information content for parameter estimation.

A challenging test problem and real-life applications show the performance of the methods.

2 Parameter estimation: Nonlinear DAE models and data

In this paper, we consider processes, which can be modeled by systems of differential-algebraic equations (DAEs)

$$\begin{aligned} \dot{y}(t) &= f(t, y(t), z(t), p, q, u(t)) \\ 0 &= g(t, y(t), z(t), p, q, u(t)) \end{aligned} \quad t \in [t_0; t_f] \quad (1)$$

Typically, e.g., if the DAE models a chemical reaction system, the DAE is of index 1 (i.e. $\partial g / \partial z$ is regular), stiff and contains specific nonlinearities. Other common application fields are biological systems (e.g. [6, 52]) and mechanical or bio-mechanical multibody system dynamics (e.g. [17] or [50]).

More general dynamics (implicit differential equations, differential equations of higher order, DAEs of higher index etc.) in principle often can be reduced to the form (1).

Moreover, the semi-discretization of a partial differential equation can lead to high-dimensional, stiff and nonlinear DAEs (e.g. [27]) that typically have special structures that can be exploited in the numerical procedures.

We denote by $t \in [t_0; t_f]$ the *time* (but it could also be a spatial direction) and by $x(t) := (y(t), z(t))$ the state variables, combining the *differential variables* $y \in \mathbb{R}^{n_y}$ and the *algebraic variables* $z \in \mathbb{R}^{n_z}$. The values of the *parameters* $p \in \mathbb{R}^{n_p}$ are unknown and have to be determined by parameter estimation. The *control functions* $u : \mathbb{R} \rightarrow \mathbb{R}^{n_u}$ and *control variables* $q \in \mathbb{R}^{n_q}$ are given in the case of parameter estimation. In case of experimental design they are the variables to be optimized in order to determine the design and the operation of experiments in such a way that the resulting measurements deliver parameter estimates with best statistical quality.

It is assumed that experiments have been carried out at the given times t_j , $j = 1, \dots, \mathcal{M}$, yielding the measurements η_{ij} , $i = 1, \dots, \mathcal{M}_j$, $j = 1, \dots, \mathcal{M}$, of the observation functions b_{ij} depending on the state variables $x(t) = (y(t), z(t))$ and the parameters p

$$\eta_{ij} = b_{ij}(t_j, x^{true}(t_j), p^{true}) + \varepsilon_{ij},$$

which are subject to measurement errors ε_{ij} . Here, p^{true} are the “true” values of the parameters and x^{true} the corresponding state variables. Note that this formulation includes the case when several model quantities are measured at a time t_j .

Now we estimate parameters by minimizing the deviation between the model and the data:

$$\| r_1(x(t_1), \dots, x(t_{\mathcal{M}}), p) \| := \left\| \begin{pmatrix} \vdots \\ (\eta_{ij} - b_{ij}(t_j, x(t_j), p)) / \sigma_{ij} \\ \vdots \end{pmatrix} \right\|$$

in an appropriate (weighted) *norm* with weights σ_{ij}^{-1} . The type of the norm is motivated by the statistical distribution of the measurement errors. If the errors are independent, normally

distributed with zero mean and known variances ($\mathcal{N}(0, \sigma_{ij}^2)$), minimizing a weighted least squares function

$$\min_{x,p} \sum_{i,j} (\eta_{ij} - b_{ij}(t_j, x(t_j), p))^2 / \sigma_{ij}^2$$

yields a *maximum likelihood estimate*, see e.g. [32, 31, 23]. Methods for unconstrained l_2 problems go back to Gauss [31]. Powerful methods and software for l_2 based parameter estimation of dynamic systems are available, as described e.g. in [14, 16, 46]. But in case of Laplace-distribution, l_1 estimation

$$\min_{x,p} \sum_{i,j} |\eta_{ij} - b_{ij}(t_j, x(t_j), p)| / \sigma_{ij}$$

yields a maximum likelihood estimate, see e.g. [13]. Further, it is well-known that l_1 parameter estimation possesses a very remarkable property, namely, that the optimal solution is typically *less sensitive to the effect of outliers* in data than, e.g., an optimal solution of l_2 based parameter estimation and therefore l_1 estimation is used for *robust* parameter estimation, see [35]. l_1 optimization problems were first studied by Boscovic [22] already in 1758.

Frequently, there is additional knowledge about the state variables and the parameters in the model, such as initial conditions, parameter bounds or boundary conditions, positivity etc. This information can be considered as equality or inequality constraints at times θ_i , $i = 1, \dots, \mathcal{K}$,

$$r_2(x(\theta_1), \dots, x(\theta_{\mathcal{K}}), p) = 0, \quad r_3(x(\theta_1), \dots, x(\theta_{\mathcal{K}}), p) \geq 0. \quad (2)$$

Summing up, the parameter estimation problem can be formulated as

$$\begin{aligned} \min_{x,p} \quad & ||r_1(x(t_1), \dots, x(t_{\mathcal{M}}), p)||, \\ \text{s.t.} \quad & (x, p) \text{ solves DAE (1),} \\ & r_2(x(\theta_1), \dots, x(\theta_{\mathcal{K}}), p) = 0, \\ & r_3(x(\theta_1), \dots, x(\theta_{\mathcal{K}}), p) \geq 0. \end{aligned} \quad (3)$$

Remarks:

- Typically, data do not only come from single experiments, but from a series of multiple experiments. This leads to large-scale problems of parameter estimation in families of boundary value problems. Methods that exploit the resulting structures efficiently are described, e.g., in [45, 46, 11, 36].
- Note that parameter estimation problems besides measurement data η_{ij} also need standard deviations σ_{ij} of the measurement error as input from the user.
- The formulation (3) of the parameter estimation problem includes in particular that also part of the state variables have to be determined and thus also covers the case of state estimation.

3 Numerical treatment of nonlinear least squares problems constrained by DAE boundary value problems

3.1 The initial value problem approach

In practice, parameter estimation problems in differential equations are often treated by simply coupling an integrator (for the numerical solution of the initial value problem in DAE) with an optimization procedure (for the solution of the least squares problem). The differential equation is solved with estimated initial values and parameters, and these estimates are iterated in order to minimize the objective function and to fulfill further constraints (see e.g. [42, 44]).

The success of this approach depends significantly on the choice of the integrator and the optimization algorithm, and their combination. Furthermore, this approach is usually only used for differential equations with explicitly given initial conditions. But even if realized in an optimal way, this approach suffers from principle disadvantages (see, e.g., [16], where also the example given in section 6 is presented):

- The parametrization of the problem by initial values and the unknown parameters is rarely suitable for the inverse character of the problem: the numerous data available for the solution are not used directly, which typically makes the procedure rather inefficient.
- For certain choices of the initial guesses for parameters and initial values, the initial value problem may be hard or impossible to solve. Severe stability problems may arise.

3.2 The boundary value problem approach

To overcome these problems the boundary value problem approach has been developed [14, 15, 7, 16, 46, 17, 50, 49] as a versatile tool for the numerical treatment of parameter estimation problems in differential equations. The basic principle consists in discretising the model (1) including possible boundary conditions like a boundary value problem – independent of its specific nature – and in treating the resulting discretized model as nonlinear constraint of the least squares objective function. This discretization is accomplished by finite differences, collocation, see e.g. [7, 1], or multiple shooting methods, see e.g. [25, 14].

In the following, we briefly describe the essence of the procedure in case of multiple shooting and a DAE model.

First one chooses a suitable grid of multiple shooting nodes τ_j

$$t_0 = \tau_0 < \tau_1 < \dots < \tau_m = t_f,$$

covering the interval, where measurements are given.

At each grid point the values of the state variables $s_j := (s_j^y, s_j^z)$ are chosen as additional unknowns and m relaxed DAE initial value problems¹

¹ Since in the parameter estimation problem control functions u and control variables q are fixed, we omit them from the problem formulation in this section for notational simplicity.

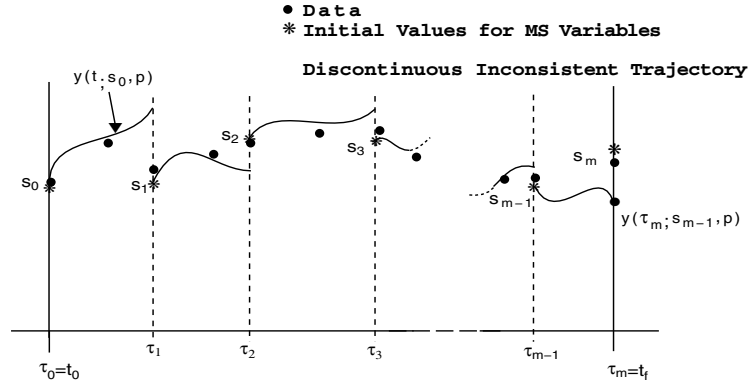


Fig. 1 Multiple shooting approach

$$\dot{y} = f(t, y, z, p), \quad (4)$$

$$0 = g(t, y, z, p) - \alpha(t)g(\tau_j, s_j^y, s_j^z, p), \quad (5)$$

$$y(\tau_j) = s_j^y, \quad z(\tau_j) = s_j^z, \quad (6)$$

are solved on each subinterval $[\tau_j, \tau_{j+1}]$ yielding a solution $x(t; s_j, p)$ for $t \in [\tau_j, \tau_{j+1}]$. The decreasing function $\alpha(t)$ ($\alpha(\tau_j) = 1, \alpha(t) \geq 0$) is chosen in order to improve consistency of the algebraic conditions in the course of the integration.

Solutions of dynamic systems, generated by this procedure, are usually neither continuous at τ_j , nor feasible for the original algebraic equation. This has to be enforced by additional matching conditions. Inserting the computed values $x(t; s_j, p)$, $\tau_j \leq t \leq \tau_{j+1}$, into problem (3) one obtains a constrained problem in the variables $(s, p) = (s_0, \dots, s_m, p)$:

$$\begin{aligned}
 \min_{s, p} \quad & \|r_1(s, p)\| \\
 \text{s.t.} \quad & r_2(s, p) = 0, \quad r_3(s, p) \geq 0, \\
 & h_j(s_j, s_{j+1}, p) := y(\tau_{j+1}; s_j, p) - s_{j+1}^y = 0, \quad j = 0, \dots, m-1, \\
 & g(\tau_j, s_j, p) = 0, \quad j = 0, \dots, m.
 \end{aligned} \quad (7)$$

Multiple shooting possesses several advantages:

1. It is possible to include a priori information about the state variables, e.g., derived from the measurements or from expert knowledge, by a proper choice of initial guesses for the additional variables s_j . Thus, it can be ensured that the initial solutions $x(t; s_j, p)$ remain close to the data. It can be shown that this usually damps the influence of poor parameter guesses.
2. The adequate choice of initial guesses for the state variables (and the application of a Gauss-Newton method for the solution of the constrained minimization problem) typically avoids convergence to local minima with large residuals.

3. The scheme is numerically stable. The splitting of the integration interval limits error propagation and allows to solve parameter estimation problems even for unstable or chaotic systems [5, 18].
4. For model problems, where the right-hand-side as well as further conditions are linear in the parameters and exact continuous measurements of the state variables are given, it can be shown that reducing the multiple shooting grid size and choosing the given measured states as initial guesses results in a reduction of nonlinearity of the parameter estimation problem. In the limit only a linear problem has to be solved (see [47], where a proof for a collocation method is presented). Also for general nonlinear problems the BVP approach typically reduces the nonlinearity of the parameter estimation problem.
5. The matching conditions induce a very specific BVP structure in the problem equations, which can be exploited in particular for the design of parallel algorithms. It allows not only for a parallel evaluation of the matching conditions and their derivatives but also for an $\mathcal{O}(m)$ algorithm for the decomposition of the linear systems in a Gauss-Newton-type optimization algorithm (see [30]).

Remark

As alternative to multiple shooting, collocation (see e.g. Ascher et al [1], Bär [7] Schulz [47, 48] or Biegler [26]) can be used for the discretization of the BVP problem. In this case the matching conditions are replaced by the collocation conditions. Collocation typically leads to constrained minimization problems of much higher dimension than in case of multiple shooting.

3.3 Gauss-Newton method

The multiple shooting parameterization of the BVP constraint yields a finite dimensional, possibly large-scale, nonlinear constrained approximation problem (7) which can be formally written as

$$\begin{aligned}
 \min_X \quad & \sum_{i \in I_1} |F_{1i}(X)|^\nu, \quad \nu = 1 \text{ or } 2, \quad I_1 := \{1, \dots, m_1\} \\
 \text{s.t.} \quad & F_{2i}(X) = 0, \quad i \in I_2 := \{1, \dots, m_2\}, \\
 & F_{3i}(X) \geq 0, \quad i \in I_3 := \{1, \dots, m_3\}.
 \end{aligned} \tag{8}$$

Note, that the equality conditions $F_{2i}(X) = 0, i \in I_2$, include the matching conditions induced by multiple shooting. We assume that the functions $\mathcal{F}_1 := [F_{1i}]_{i \in I_1} : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^{m_1}$, $\mathcal{F}_2 := [F_{2i}]_{i \in I_2} : D \subset \mathbb{R}^n \rightarrow \mathbb{R}^{m_2}$, and $\mathcal{F}_3 := [F_{3i}]_{i \in I_3}$ are twice-continuously differentiable. The number of variables in problem (8) is equal to the number of differential and algebraic variables multiplied by the number of multiple shooting nodes plus the number of parameters.

We apply generalized Gauss-Newton methods to solve problem (8) using a tailored linear algebra to exploit the special structures arising from the multiple shooting discretization. The iteration scheme is (basically) given by

$$X^{k+1} = X^k + t^k \Delta X^k, \quad 0 < t^k \leq 1, \tag{9}$$

where the increment ΔX^k is the solution of the following linear constrained problem at $X = X^k$

$$\begin{aligned} \min_{\Delta X \in \mathbb{R}^n} \quad & \sum_{i \in I_1} |F_{1i}(X) + J_{1i}(X)\Delta X|^\nu, \quad \nu = 1 \text{ or } 2, \\ \text{s.t.} \quad & F_{2i}(X) + J_{2i}(X)\Delta X = 0, i \in I_2, \quad F_{3i}(X) + J_{3i}(X)\Delta X \geq 0, i \in I_3. \end{aligned} \quad (10)$$

Here $J_{ki}(X)$ denotes the gradient of $F_{ki}(X)$, that is the row-vector $\nabla F_{ki}(X) = (\frac{\partial F_{ki}}{\partial X})$.

3.4 Convergence properties

In this subsection we discuss the convergence properties of the Gauss-Newton-method ((9), (10)). First we consider the l_1 parameter estimation and later on the l_2 case.

Let us start with recalling the optimality conditions for the linear l_1 problem ((8), $\nu = 1$). We define the active set $I(X)$ at the point X as

$$I(X) = I_1(X) \cup I_2(X) \cup I_3(X), \quad I_k(X) = \{i \in I_k : F_{ki}(X) = 0\}, k = 1, 2, 3,$$

and define a *regular point* X as a point for which a *constraint qualification* holds, that is the Jacobian of all active constraints and zero components of the cost function at X has full row rank

$$\text{rank } \mathcal{J}_0(X) = m_0 := |I(X)|, \quad \mathcal{J}_0(X) := [\mathcal{J}_{ki}(X)]_{i \in I_k(X), k=1,2,3}. \quad (11)$$

Theorem 3.1 (Necessary conditions) *Let X^* be a regular solution of problem (8). Then there is a unique vector of Lagrange multipliers $\lambda^* = [\lambda_k^*]_{k=1,2,3}$, $\lambda_k^* \in \mathbb{R}^{m_k}$, $k = 1, 2, 3$, such that the following conditions (the first-order necessary optimality conditions) are true for the pair (X^*, λ^*) :*

$$\begin{aligned} \sum_{k=1,2,3} \lambda_k^{*T} J_k(X^*) &= 0, \\ |\lambda_{1i}^*| &\leq 1, \quad i \in I_1(X^*), \quad \lambda_{1i}^* = -\text{sign} F_{1i}(X^*), \quad i \in I_1 \setminus I_1(X^*), \\ \lambda_{3i}^* &\geq 0, \quad i \in I_3(X^*), \quad \lambda_{3i}^* = 0, \quad i \in I_3 \setminus I_3(X^*). \end{aligned} \quad (12)$$

Further, the point (X^*, λ^*) satisfies the second-order necessary optimality conditions:

$$d^T H d \geq 0, \quad H = H(X^*, \lambda^*) := - \sum_{k=1,2,3} \sum_{i \in I_k} \lambda_{ki}^* \frac{\partial^2 F_{ki}(X^*)}{\partial X^2}$$

for all directions d from $P(X^*, \lambda^*)$

$$\begin{aligned} P(X, \lambda) = \{d \neq 0 \mid & J_{2i}(X)d = 0, \quad i \in I_2; \\ & J_{1i}(X)d = 0, \quad \text{if } |\lambda_{1i}| < 1, \\ & \lambda_{1i} J_{1i}(X)d \leq 0, \quad \text{if } \lambda_{1i} = 1 \text{ or } -1, \quad i \in I_1(X); \\ & J_{3i}(X)d = 0, \quad \text{if } \lambda_i > 0, \\ & J_{3i}(X)d \geq 0, \quad \text{if } \lambda_i = 0, \quad i \in I_3(X)\}. \end{aligned}$$

Theorem 3.2 (Sufficient optimality conditions) Let (X^*, λ^*) satisfy the first-order necessary optimality conditions (12) and the matrix H is positive definite for all directions $d \in P(X^*, \lambda^*)$, that is

$$d^T H d > 0, \quad \forall d \in P(X^*, \lambda^*). \quad (13)$$

Then X^* is a strict local minimum of (8).

Corollary 3.3 Let (X^*, λ^*) satisfy the first-order necessary conditions (12) and strict complementarity

$$|\lambda_{1i}| < 1, i \in I_1(X), \quad \lambda_{3i} > 0, i \in I_3(X). \quad (14)$$

and let the Hessian H be positive definite for all $d \in Z(X^*)$,

$$Z(X) := \{d \neq 0 \mid J_{ki}(X)d = 0, i \in I_k(X), k = 1, 2, 3\}.$$

Then X^* is a strict local minimum of (8).

Proof follows from applying optimality conditions of nonlinear programming to the following problem, which is equivalent to problem (8)

$$\begin{aligned} \min_{X, \chi} \quad & \sum_{i \in I_1} \chi_i \\ \text{s.t.} \quad & \chi_i + F_{1i}(X) \geq 0, \quad \chi_i - F_{1i}(X) \geq 0, \quad \chi_i \geq 0, \quad i \in I_1, \\ & F_{2i}(X) = 0, i \in I_2, \quad F_{3i}(X) \geq 0, i \in I_3. \end{aligned}$$

Applying the optimality conditions to the linear l_1 problem (10), which read

$$\begin{aligned} \sum_{k=1,2,3} \lambda_{k,LP}^* J_k(X) &= 0, \\ |\lambda_{1,i,LP}^*| &\leq 1, i \in I_1(\Delta X^*), \\ \lambda_{1,i,LP}^* &= -\text{sign}(F_{1i}(X) + J_{1i}(X)\Delta X^*), i \in I_1 \setminus I_1(\Delta X^*), \\ \lambda_{3,i,LP}^* &\geq 0, i \in I_3(\Delta X^*), \quad \lambda_{3,i,LP}^* = 0, i \in I_3 \setminus I_3(\Delta X^*), \end{aligned} \quad (15)$$

we may conclude that its solution is defined by a generalized inverse. Denote

$$\mathcal{F}(X) = [F_{ki}(X)]_{i \in I_k, k=1,2,3}, \quad \mathcal{J}(X) = [J_{ki}(X)]_{i \in I_k, k=1,2,3}.$$

Theorem 3.4 (Existence of a generalized inverse) Let ΔX^* solve problem (10). Assume further that

$$\text{rank } \mathcal{J}_0 = |I(\Delta X^*)|, \quad \mathcal{J}_0 = \mathcal{J}_0(\Delta X, X) := [J_{ki}(X)]_{i \in I_k(\Delta X), k=1,2,3} \quad (16)$$

and that there exist Lagrange multipliers λ_{LP}^* satisfying the optimality conditions (15) for (10), such that $\text{rank } \mathcal{J}_0 = n$ and strict complementarity holds. Then

i) $(\Delta X^*, \lambda_{LP}^*)$ is a unique Kuhn-Tucker point of the linear l_1 problem (10), and ΔX^* is a strict minimum.

ii) There is a linear operator $\mathcal{J}^+ : R^{m_1+m_2+m_3} \rightarrow R^n$, such that

$$\Delta X^* = -\mathcal{J}(X)^+ \mathcal{F}(X).$$

iii) The solution operator is a generalized inverse, that is, it possesses the property

$$\mathcal{J}(X)^+ \mathcal{J}(X) \mathcal{J}(X)^+ = \mathcal{J}(X)^+,$$

and up to column permutation is explicitly given by

$$\mathcal{P} \mathcal{J}(X)^+ = [0, \mathcal{J}_0^{-1}(X)].$$

Here $\mathcal{P}$ is a permutation matrix.

Analyzing further the optimality conditions, we may conclude that a solution X^* of the nonlinear l_1 problem ((8), $\nu = 1$), is characterized by an optimal “active set” which contains information about active inequality constraints and zero components of the cost function. The following result shows that the Gauss-Newton method ((9), (10)) eventually identifies the optimal “active set” of the nonlinear l_1 problem (8).

Theorem 3.5 Suppose that X^* is a solution of the problem (8) that satisfies the constraint qualification (11) and $n = m_0$, that is exactly n components of the cost function and constraints in the nonlinear l_1 problem are zero. Assume further that strict complementarity holds. Then, if (X^k, λ^k) is sufficiently close to (X^*, λ^*) , the linear l_1 problem (10) has a unique solution $(\Delta X^k, \lambda_{LP}^k)$, whose active set $I(\Delta X^k)$ is the same as the active set $I(X^*)$ of the nonlinear problem (8) at X^* .

Summing up, the iterations of the Gauss-Newton method ((9), (10)) are attracted by one of the local minima satisfying the regularity assumptions mentioned above. Neither the active sets in linear problems (10), constructed at X^k , will change, nor non-zero components of the cost function and the constraints will change their signs as the iterates X^k generated by the generalized Gauss-Newton method ((9), (10)) approach such a minimizer X^* . Moreover, these sets coincide with the corresponding optimal sets at X^* . Hence, the full-step ($t^k \equiv 1$) method becomes equivalent to the Newton method applied for solving a system of nonlinear equations

$$\mathcal{F}_{0,*}(X) = 0, \quad \mathcal{F}_{0,*}(X) := \begin{bmatrix} F_{k,i}(X) \end{bmatrix}_{i \in I_k(X^*), k=1,2,3} \quad (17)$$

and thus has a quadratic rate of local convergence.

The equations (17) include the equality constraints, the active inequality constraints and the zero components of the cost function at the solution, so that we can ignore non-active constraints and non-zero components of the cost function. Consequently, the solution X^* interpolates $n - m_0$ “best” measurements, which means that X^* is less sensitive to outliers.

Global Convergence Choosing the step length t^k by means of classical line search methods based on the exact penalty function

$$T_1(X) := \sum_{i \in I_1} |F_{1i}(X)| + \sum_{i \in I_2} \alpha_i |F_{2i}(X)| + \sum_{i \in I_3} \beta_i \max\{0, -F_{3i}(X)\}$$

with sufficiently large weights $\alpha_i > 0$, $i = 1, \dots, m_2$, $\beta_i > 0$, $i = 1, \dots, m_3$, ensures global convergence. However, it is well known that already in mildly ill-conditioned problems such a step size strategy may be very inefficient, since it may produce very small step sizes. Therefore we use the “restrictive monotonicity test”, see [16], [19], that has proved to be very effective

in practical applications. We interpret the Gauss-Newton method for l_1 parameter estimation problem as Newton's method applied to the nonlinear system of the active constraints and the zero components of the cost function (17) and use the restrictive monotonicity test for the "natural level function"

$$T_k(X^k + t\Delta X^k) = \|A\mathcal{F}_{0,k}(X^k + t\Delta X^k)\|_2, \quad A = \mathcal{J}_{0,k}^{-1}(X^k).$$

The idea of the test is that at X^k we consider the relaxed system of nonlinear equations

$$\mathcal{F}_{0,k}(X) - (1-t)\mathcal{F}_{0,k}(X^k) = 0, \quad 0 \leq t \leq 1, \quad (18)$$

and choose the maximal step size $t \in]0, 1]$ for which the iterates of the simplified Newton method applied to (18) contract. The test may be also interpreted as following the Newton path. Note, however, that the active inequality constraints incorporated in $\mathcal{J}_{0,k}(X^k)$ as well as the zero components of the cost function are changing along the solution of the problem, so that the step-size strategy is combined with an additional monitoring of the changing active sets. The restrictive monotonicity test has shown very good performance in practice; for a theoretical justification of the test we refer to [19].

l_2 Gauss-Newton Method Here we summarize the convergence properties of the l_2 based Gauss-Newton method. The convergence analysis is based on the optimality conditions for the nonlinear parameter estimation problem ((8), $\nu = 2$). First, we define the active set $I_C(X)$ at the point X as $I_C(X) = I_2(X) \cup I_3(X)$, and define a *regular* point X as a point for which the *constraint qualification* holds, that is the Jacobian of all active constraints at X has full row rank

$$\text{rank } \mathcal{J}_C(X) = m_C := |I_C(X)|, \quad \mathcal{J}_C(X) := [\mathcal{J}_{ki}(X)]_{i \in I_k(X), k=1,2}. \quad (19)$$

With the help of the Lagrange function

$$L(X, \tilde{\lambda}) = \frac{1}{2} \|\mathcal{F}_1(X)\|_2^2 - \sum_{k=1,2} \tilde{\lambda}_k^T \mathcal{F}_k(X),$$

where $\tilde{\lambda} = [\tilde{\lambda}_k]_{k=1,2}$, $\tilde{\lambda}_k \in R^{m_k}$ are Lagrange multipliers, we may formulate

Theorem 3.6 (Necessary conditions) *Let X^* be a regular solution of problem ((8), $\nu = 2$). Then there is a unique vector of Lagrange multipliers $\tilde{\lambda}^*$, such that the following conditions (first-order necessary optimality conditions) are true for $(X^*, \tilde{\lambda}^*)$:*

$$\begin{aligned} \frac{\partial}{\partial X} L(X^*, \tilde{\lambda}^*) &= \mathcal{F}_1(X^*)^T \mathcal{J}_1(X^*) - \sum_{k=1,2} \tilde{\lambda}_k^{*T} \mathcal{J}_k(X^*) = 0, \\ \tilde{\lambda}_{3i}^* &\geq 0, i \in I_3(X^*), \quad \tilde{\lambda}_{3i}^* = 0, i \in I_3 \setminus I_3(X^*). \end{aligned} \quad (20)$$

Further, the point $(X^*, \tilde{\lambda}^*)$ satisfies the second-order necessary optimality conditions:

$$\begin{aligned} d^T H d &\geq 0, \quad H = H(X^*, \tilde{\lambda}^*) := \frac{\partial^2}{\partial X^2} L(X^*, \tilde{\lambda}^*) = \\ &\mathcal{J}_1(X^*)^T \mathcal{J}_1(X^*) + \sum_{i \in I_1} F_{1i}(X^*) \frac{\partial J_{1i}(X^*)}{\partial X} - \sum_{k=1,2} \sum_{i \in I_k} \tilde{\lambda}_{ki}^* \frac{\partial^2 F_{ki}(X^*)}{\partial X^2} \end{aligned}$$

for all directions $d \in P(X^*, \tilde{\lambda}^*)$:

$$P(X, \tilde{\lambda}) = \{d \neq 0 \mid \begin{array}{ll} J_{2i}(X)d = 0, & i \in I_2(X), \\ J_{3i}(X)d = 0, & \text{if } \tilde{\lambda}_3 > 0, \\ J_{3i}(X)d \geq 0, & \text{if } \tilde{\lambda}_{3i} = 0, \end{array} i \in I_3(X)\}$$

Theorem 3.7 (Sufficient optimality conditions) Let $(X^*, \tilde{\lambda}^*)$ satisfy the first-order necessary optimality conditions (20) and assume that for all directions $d \in P(X^*, \tilde{\lambda}^*)$ the matrix H is positive definite. Then X^* is a strict local minimum of ((8), $\nu = 2$).

Corollary 3.8 Let $(X^*, \tilde{\lambda}^*)$ satisfy the first-order necessary conditions (20), strict complementarity $\tilde{\lambda}_{3i} > 0, i \in I_3(X)$, and let the Hessian H be positive definite for all directions $d \in \{d \neq 0 \mid \mathcal{J}_C(X)d = 0\}$. Then X^* is a strict local minimum of ((8), $\nu = 2$).

Corollary 3.8 immediately implies that, if X^* satisfies the constraint qualification (19) and the conditions of 3.8, then X^* is a fixed point of the Gauss-Newton iterations. Further we may conclude that as the iterates of the Gauss-Newton method approach a local minimum satisfying constraint qualification (19) and the conditions of 3.8, namely positive definiteness and strict complementarity, the active set remains fixed:

Theorem 3.9 Suppose that X^* is a solution of the problem ((8), $\nu = 2$), that satisfies the constraint qualification (19) and the assumptions of Corollary 3.8. Then if (X^k, λ^k) is sufficiently close to (X^*, λ^*) , the linear l_2 problem (10) has a unique solution $\Delta X^k, \lambda_{LP}^k$, whose active set $I(\Delta X^k)$ is the same as the active set $I(X^*)$ of the nonlinear problem (8) at X^* .

This means that in a neighbourhood D of the solution X^* satisfying the conditions of Theorem 3.9 we may ignore inequalities not belonging to $I_C(X^*)$ and consider (locally) only equality constrained problems. That is why we finish this section with several results for the Gauss-Newton method applied to the solution of equality constrained problems

$$\min_{X \in \mathbb{R}^n} \frac{1}{2} \sum_{i \in I_1} F_{1i}^2(X), \quad \text{s.t. } F_{Ci}(X) = 0, \quad i \in I_C, I_C = \{1, \dots, m_C\}. \quad (21)$$

First of all, if the Jacobians $\mathcal{J}_1(X) = [J_i(X)]_{i \in I_1}$ and $\mathcal{J}_C(X) = [J_i(X)]_{i \in I_C}$ satisfy two regularity assumptions on the domain D

$$\text{rank } \mathcal{J}_C(X) = m_C, \text{ rank } \mathcal{J}(X) = n, \mathcal{J}(X) = \begin{bmatrix} \mathcal{J}_1(X) \\ \mathcal{J}_C(X) \end{bmatrix}, \quad (22)$$

then the linearized problem

$$\begin{aligned} \min_{\Delta X \in \mathbb{R}^n} \quad & \frac{1}{2} \sum_{i \in I_1} (F_{1i}(X) + J_{1i}(X)\Delta X)^2, \\ \text{s.t.} \quad & F_{Ci}(X) + J_{Ci}(X)\Delta X = 0, \quad i \in I_C, \end{aligned} \quad (23)$$

has a unique solution ΔX^k and a unique Lagrange vector $\tilde{\lambda}^k$ satisfying the following optimality conditions

$$\begin{aligned} \mathcal{J}_1^T(X)\mathcal{J}_1(X)\Delta X^k + \mathcal{J}_C^T(X)\lambda^k &= -\mathcal{J}_1^T(X)\mathcal{F}_1(X), \\ \mathcal{J}_C(X)\Delta X^k &= -\mathcal{F}_C(X). \end{aligned} \quad (24)$$

Here, $\mathcal{F}_1(X) = [F_i(X)]_{i \in I_1}$ and $\mathcal{F}_C(X) = [F_i(X)]_{i \in I_C}$. Using (24) one can easily show that under the regularity conditions (22) ΔX^k can be formally written with the help of a solution operator $\mathcal{J}^+$

$$\Delta X^k = -\mathcal{J}^+(X^k)\mathcal{F}(X^k).$$

As in the l_1 -case the solution operator $\mathcal{J}^+$ is a generalized inverse, that is, it satisfies the defining conditions

$$\mathcal{J}^+ \mathcal{J} \mathcal{J}^+ = \mathcal{J}^+,$$

and in the l_2 -case is explicitly given by

$$\mathcal{J}^+(X) = \begin{pmatrix} \mathcal{I} & 0 \end{pmatrix} \begin{pmatrix} \mathcal{J}_1^T(X)\mathcal{J}_1(X) & \mathcal{J}_C(X)^T \\ \mathcal{J}_C(X) & 0 \end{pmatrix}^{-1} \begin{pmatrix} \mathcal{J}_1(X)^T & 0 \\ 0 & \mathcal{I} \end{pmatrix}. \quad (25)$$

The local convergence properties of the Gauss-Newton method for equality constrained l_2 problems (21) may be formulated as follows (see [16]).

Theorem 3.10 (Local Contraction Theorem) *Let D be the domain defined in Theorem 3.9. Assume further that $\mathcal{J}^+(X)$, $X \in D$, satisfies the following conditions for all $t \in]0, 1]$, $X, Z, Y = X + \Delta X \in D$ with $\Delta X = -J(X)^+ F(X) \neq 0$:*

$$\|\mathcal{J}(Y)^+ (\mathcal{J}(X + t\Delta X) - \mathcal{J}(X)) \Delta X\| \leq \omega t \|\Delta X\|^2, \quad (26)$$

$$\omega < \infty,$$

i.e. a global bound ω for the “curvature” exists, and

$$\|(\mathcal{J}(Z)^+ - \mathcal{J}(X)^+)R(X)\| \leq \kappa(X)\|Z - X\| < \kappa\|Z - X\|$$

where $R(X) := \mathcal{F}(X) - \mathcal{J}(X)\mathcal{J}(X)^+\mathcal{F}(X)$ (residuals) with $\kappa < 1$.

Assume further that the initial guess X^0 is sufficiently close to a solution:

$$\delta^0 := \frac{\omega}{2} \|\Delta X^0\| < 1. \quad (27)$$

Then the following holds:

- *if $D^0 := B(X^0, \|\Delta X^0\|/(1 - \delta^0)) \subset D$, then the sequence of full-step iterates defined by $((9), (t^k = 1))$ remains in D^0 ,*
- *there exists $X^* \in D^0$ with*

$$J(X^*)^+ F(X^*) = 0 \quad (28)$$

and $X^k \rightarrow X^$ ($k \rightarrow \infty$),*

- *an a priori error estimate holds*

$$\|X^k - X^*\| \leq (\delta^0)^k \frac{\|\Delta X^0\|}{1 - \delta^0},$$

- *and the convergence is linear with*

$$\|\Delta X^{k+1}\| \leq \left(\frac{\omega}{2} \|\Delta X^k\| + \kappa \right) \|\Delta X^k\|.$$

3.5 “Statistically” stable and unstable solutions

Now let us discuss further properties of the l_1 based Gauss-Newton method. We saw that the generalized Gauss-Newton method ((9), (10)) can be interpreted as a Sequential Linear Programming Method (SLP) in the l_1 case since we ignore the information about second-order derivatives. As a SLP method the generalized Gauss-Newton method for (8) has several advantages. It does not use second order derivative information and the local problems are linear programming problems. Under certain regularity assumptions about the solution, the method shows quadratic rate of local convergence. Moreover, these solutions are strict local minima, and according to the implicit function theorem applied to the optimality conditions (12) such a solution depends continuously and differentiable on (small) perturbations of the minimization problem, e.g. small perturbations of measurements. There are problems, however, for which the Gauss-Newton method may have rather slow local convergence rate or may even fail. The reason is that the linear model (10), which forms the basis of the Gauss-Newton method, is an inadequate local representation of such nonlinear problems, where the second-order information cannot be ignored. Similar situations appear in applying the l_2 based Gauss-Newton method to parameter estimation problems with large residuals, see e.g. [16] for a thorough analysis (including constrained l_2 parameter estimation problems). In particular, it is shown there that, if the full-step l_2 based Gauss-Newton method is not attracted by a minimum, then this minimum does not satisfy a criterion for statistical stability. Using SQP-type methods for the nonlinear constrained l_1 optimization problem one could force convergence to such a solution even in such a case, see e.g. [41, 34] for unconstrained l_1 problems and [40] for constrained l_1 problems. However, as parameter estimates such solutions are undesirable in the following sense. Recall, that we are interested not just in the solution of an optimization problem, but rather in getting reliable, trustworthy estimates for the parameters. We want to get estimates that are stable against statistical perturbations in the measurement errors.

We will show that a solution that does not satisfy the assumptions, which guarantee uniqueness of primal and dual solutions in the linear problems in its neighbourhood, even if it is a strict minimum of the nonlinear constrained l_1 problem (8), cannot be expected to be a continuous deformation of the “true” parameter values under statistical perturbations due to measurement errors.

Theorem 3.11 *Let X^* be a local minimizer of the equality constrained l_1 nonlinear problem*

$$\begin{aligned} \min_{X \in \mathbb{R}^n} \quad & \sum_{i \in I_1} |F_{1i}(X)|, \\ \text{s.t.} \quad & F_{2i}(X) = 0, \quad i \in I_2, \end{aligned} \quad (29)$$

satisfying the constraint qualification (11), $|I(X^)| = n$ and the strict complementarity (14) of the corresponding λ^* . Then X^* is a strict local minimizer for any of the following τ -dependent problems*

$$\begin{aligned} \min_{X \in \mathbb{R}^n} \quad & \sum_{i \in I_1} |F_{1i}(X, \tau)|, \\ \text{s.t.} \quad & F_{2i}(X, \tau) = 0, \quad i \in I_2, \end{aligned} \quad (30)$$

where $F_{ki}(X, \tau) = F_{ki}(X) + (\tau - 1)F_{ki}(X^)$, $i \in I_k, k = 1, 2$, $-1 \leq \tau \leq 1, \tau \neq 0$.*

Proof. Under the assumptions of the theorem there is a unique vector of Lagrange multipliers λ^* such that

$$\sum_{k=1,2} \sum_{i \in I_k} \lambda_{ki}^* J_{ki}(X^*) = 0, \quad |\lambda_{1i}^*| \leq 1, \quad \lambda_{1i}^* = -\text{sign}(F_{1i}(X^*)), \quad i \in I_1 \setminus I_1(X^*). \quad (31)$$

Moreover, X^* is a strict local minimizer, since $P(X^*, \lambda^*) = \emptyset$. Consider now problem (30). Obviously,

$$\begin{aligned} J_{ki}(X, \tau) &= J_{ki}(X), \quad F_{ki}(X^*, \tau) = \tau F_{ki}(X^*), \quad i \in I_k, k = 1, 2, \\ F_{ki}(X^*, \tau) &= 0, \quad i \in I_k(X^*), \quad k = 1, 2, \forall \tau, \tau \neq 0, \\ \text{sign}(F_{1i}(X^*, \tau)) &= \text{sign}(F_{1i}(X^*)), \quad \tau > 0, \\ \text{sign}(F_{1i}(X^*, \tau)) &= -\text{sign}(F_{1i}(X^*)), \quad \tau < 0, \quad i \in I_1 \setminus I_1(X^*). \end{aligned}$$

Then the pair $(X^*, \lambda^*(\tau))$ satisfies the first-order necessary conditions (31), if we choose

$$\lambda^*(\tau) = \lambda^*, \quad \text{for } \tau > 0, \quad \lambda^*(\tau) = -\lambda^*, \quad \text{for } \tau < 0. \quad (32)$$

Further, $P(X^*, \lambda^*, \tau) = P(X^*, \lambda^*) = \emptyset$, and, consequently, X^* is a strict local minimizer in problem (30) for any $\tau \neq 0$. $\diamond$

Theorem 3.12 *Let X^* be a regular strict local minimizer of the equality constrained l_1 problem (29). Assume further that $P(X^*, \lambda^*) \neq \emptyset$. Then*

- X^* is a stationary point for any τ in problem (30);
- X^* is not a minimizer for any $\tau < 0$ in problem (30).

Proof. It follows from the proof of the previous theorem that $(X^*, \lambda^*(\tau))$ satisfies the optimality conditions (31) with the special choice of the Lagrange multipliers (32). The Hessian $H(\tau)$ of the Lagrangian in problem (30) at the point $(X^*, \lambda^*(\tau))$ is given by

$$\begin{aligned} H(\tau) &= H(X^*, \lambda^*(\tau)) = - \sum_{k=1,2} \sum_{i \in I_k} \lambda_{ki}^*(\tau) \frac{\partial^2 F_{ki}(X^*, \tau)}{\partial X^2} \\ &= \begin{cases} H(X^*, \lambda^*) & \text{if } \tau > 0; \\ -H(X^*, \lambda^*) & \text{if } \tau < 0. \end{cases} \end{aligned}$$

$\diamond$

What can we conclude from these theorems for parameter estimation problems? Consider just for a moment an unconstrained l_1 parameter estimation problem in the form

$$\min_p \sum_{i \in \mathcal{I}} |f_i(p)| := \sum_i |Q_i(p) - \eta_i| := \sum_i |Q_i(p) - (Q_i(p^{true}) + \varepsilon_i)|, \quad (33)$$

where $Q_i(p)$ denotes a model response at some point i , p denotes unknown parameters, η_i denotes a measurement, which is assumed to be equal to the model response for the true values of the parameters subject to a measurement error ε_i , $i \in \mathcal{I}$. Let the strict local minimum

$p^*(\varepsilon)$, $\varepsilon = (\varepsilon_i, i \in \mathcal{I})$, of problem (33) be a good estimate of the true parameter values $p^{true} = p^*(0) : p^*(\varepsilon) \approx p^{true}$. Then

$$f_i(p^*(\varepsilon)) = Q_i(p^*(\varepsilon)) - Q_i(p^{true}) - \varepsilon_i \approx -\varepsilon_i.$$

Assume further that $p^*(\varepsilon)$ is a solution, which satisfies the conditions of Theorem 3.12, and consider the functions $f_i(p, \tau) := f_i(p) + (\tau - 1)f_i(p^*(\varepsilon))$, $i \in \mathcal{I}$. Then for $\tau = 1$ we have the old problem (33) with $f_i(p) = Q_i(p) - \eta_i$, the measurements $\eta_i = Q_i(p^{true}) + \varepsilon_i$ and the measurement errors ε_i , $i \in \mathcal{I}$. If $\tau = -1$, we have a new problem (33) with $\tilde{f}_i(p) = Q_i(p) - \tilde{\eta}_i$, the measurements $\tilde{\eta}_i = Q_i(p^*(\varepsilon)) + f_i(p^*(\varepsilon)) \approx Q_i(p^{true}) - \varepsilon_i$ and the measurement errors $-\varepsilon_i$, $i \in \mathcal{I}$, which means that we just inverted the signs of the measurement errors. According to Theorem 3.12, $p^*(-\varepsilon) \approx p^{true}$ is still a stationary point in the new parameter estimation problem, but not a solution anymore, although the measurement errors did not change their statistical properties. Thus, we can estimate the true values of parameters p^{true} for the errors ε , and cannot for $-\varepsilon$. We have a bifurcation: $p^*(-\varepsilon)$ jumps from the saddle point p^{true} to another point! This means that the point $p^*(-\varepsilon)$ is not a continuously differentiable estimate of $p^{true} = p^*(0)$.

Similar to the l_2 case dealt in [16] we have shown that all minima requiring second-order information and which, for this reason, are not attractive for the Gauss-Newton method, are statistically unstable: they are not estimates, i.e. they cannot be considered to be continuous deformations of the true parameter values due to perturbations by the measurement errors. From this point of view quasi-Newton methods have only limited use for parameter estimation, since they do not deliver estimates for the true parameter values! Thus, slow local convergence or no convergence of the full-step Gauss-Newton method, seen often as a disadvantage of the method, indicates deficiencies in the model or lack of data and can be considered as its *advantage*!

3.6 Evaluation of functions and Jacobians

In the course of the Gauss-Newton method the entries in the objective function and the constraints and their derivatives must be evaluated frequently. In multiple shooting, the main computational effort arises in the solution of the initial value problems (4 - 6) and the computation of the solution derivatives with respect to the unknowns. Efficient error controlled numerical integration methods that also deliver derivatives of the solution are required.

We want to mention two examples of such integrators. For the treatment of relaxed stiff DAE of index 1 in particular, the integrator DAESOL [8, 9] has been developed, a BDF method with variable mesh and order formulas based on Newton interpolation. It uses true variable mesh error estimates for order and stepsize control, and an efficient nonlinear implicit system treatment, which employs strategies developed for continuation problems [16].

For mechanical multibody systems MBSSIM [51] was developed. It treats index-reduced mechanical DAE with invariants with an advanced non-stiff multistep method.

The calculation of derivatives is not only the most time consuming part of a multiple shooting method. A perturbation analysis of constrained least-squares BVPs shows that errors in the derivative not only slow down convergence, but also produce a solution error (cf. Theorem 3.10 (28)). Therefore, we use "Internal Numerical Differentiation" (IND) procedures, which compute derivatives of the internally generated discretization schemes [16]. IND is stable in the sense of backward analysis (including discontinuity treatment and the low tolerance case).

It is accurate and allows derivative error control, and it is less expensive – computing time gains of up to 80% over usual forward differences are achieved. One of the unique features responsible for the fast performance of the multiple shooting method – the adaptive accuracy strategy [16], which keeps integration tolerances below two decimals for the most part – depends strongly on the use of IND.

3.7 Solution of linear constrained problems

In this section we describe one of the structure exploiting methods for solving linear constrained l_ν problems (10), which occur in each iteration of the Gauss-Newton method.

3.7.1 Condensing

Block- LQ decomposition Linearization of the discretized parameter estimation BVP (7) yields a large, but sparse Jacobian $\mathcal{J}$ with a special structure, which allows very effective recursive block decompositions. We describe here an LQ -decomposition, which is preferable due to its numerical stability. Here L is a lower triangular and Q is an orthogonal matrix respectively. Not only for the sake of simplicity, but rather for improving stability properties of the decomposition we handle the parameters p as constant state variables (with derivative zero) and include them in the differential state variables s_j^y . The Jacobian under consideration has the form (for the sake of simplicity we omit the point X^k and the index k):

$$\mathcal{J} = \begin{pmatrix} D_1^0 & D_1^1 & \cdots & D_1^m \\ D_2^0 & D_2^1 & \cdots & D_2^m \\ D_3^0 & D_3^1 & \cdots & D_3^m \\ D_4^0 & D_4^1 & \cdots & D_4^m \\ G^0 & H^0 & & \\ & \ddots & \ddots & 0 \\ & & \ddots & \ddots \\ & 0 & & \ddots & \ddots \\ & & & G^{m-1} & H^{m-1} \end{pmatrix}, \quad \mathcal{F} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ g_0 \\ \tilde{h}_0 \\ \tilde{h}_1 \\ \vdots \\ \vdots \\ \vdots \\ \tilde{h}_{m-1} \end{pmatrix}$$

where

$$\begin{aligned} D_i^j &:= \partial r_i / \partial (s_j), \quad j = 0, \dots, m, \quad i = 1, 2, 3 \\ D_4^0 &:= \partial g(\tau_0, s_0) / \partial (s_0), \quad D_4^j := 0, \quad j = 1, \dots, m \\ G^j &:= \begin{pmatrix} \partial y(\tau_{j+1}; s_j) / \partial (s_j) \\ 0 \end{pmatrix}, \\ H^j &:= \begin{pmatrix} -\mathcal{I} & 0 \\ \partial g(\tau_{j+1}, s_{j+1}) / \partial (s_{j+1}^y) & \partial g(\tau_{j+1}, s_{j+1}) / \partial (s_{j+1}^z) \end{pmatrix} \\ g_0 &= g(\tau_0, s_0) \\ \tilde{h}_j &= \begin{pmatrix} h_j = y(\tau_{j+1}; s_j) - s_{j+1}^y \\ g(\tau_{j+1}, s_{j+1}) \end{pmatrix}, \quad j = 0, \dots, m-1 \end{aligned} \tag{34}$$

Note that all matrices H^j have full rank, because the DAE is of index 1.

In the first step we compute an LQ -decomposition of the block $[G^0, H^0]$

$$[G^0, H^0] = [L^0, 0]Q^0,$$

with Q^0 orthogonal and L^0 lower triangular and compute the necessary changes in the corresponding blocks:

$$[0, G^1] = [T^0, \tilde{G}^1]Q^0, \quad \begin{bmatrix} D_1^0 & D_1^1 \\ D_2^0 & D_2^1 \\ D_3^0 & D_3^1 \\ D_4^0 & D_4^1 \end{bmatrix} = \begin{bmatrix} \tilde{D}_1^0 & \hat{D}_1^1 \\ \tilde{D}_2^0 & \hat{D}_2^1 \\ \tilde{D}_3^0 & \hat{D}_3^1 \\ \tilde{D}_4^0 & \hat{D}_4^1 \end{bmatrix} Q^0$$

For the next step of decomposition the matrix is now given by

$$\mathcal{J}^1 = \begin{pmatrix} \tilde{D}_1^0 & \hat{D}_1^1 & \dots & D_1^m \\ \tilde{D}_2^0 & \hat{D}_2^1 & \dots & D_2^m \\ \tilde{D}_3^0 & \hat{D}_3^1 & \dots & D_3^m \\ \tilde{D}_4^0 & \hat{D}_4^1 & \dots & D_4^m \\ L^0 & 0 & \dots & 0 \\ T^0 & \tilde{G}^1 & H^1 & 0 & \dots & 0 \\ 0 & 0 & G^2 & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & G^{m-1} & H^{m-1} \end{pmatrix}$$

In this step we compute an LQ -decomposition of the block $[\tilde{G}^1, H^1]$

$$[\tilde{G}^1, H^1] = [L^1, 0]Q^1$$

and again compute the necessary changes in the corresponding blocks

$$[0, G^2] = [T^1, \tilde{G}^2]Q^1, \quad \begin{bmatrix} \hat{D}_1^1 & D_1^2 \\ \hat{D}_2^1 & D_2^2 \\ \hat{D}_3^1 & D_3^2 \\ \hat{D}_4^1 & D_4^2 \end{bmatrix} = \begin{bmatrix} \tilde{D}_1^1 & \hat{D}_1^2 \\ \tilde{D}_2^1 & \hat{D}_2^2 \\ \tilde{D}_3^1 & \hat{D}_3^2 \\ \tilde{D}_4^1 & \hat{D}_4^2 \end{bmatrix} Q^1$$

We proceed with this procedure until the last multiple shooting block is processed:

$$[\tilde{G}^{m-1}, H^{m-1}] = [L^{m-1}, 0]Q^{m-1}; \quad \begin{bmatrix} \hat{D}_1^{m-1} & D_1^m \\ \hat{D}_2^{m-1} & D_2^m \\ \hat{D}_3^{m-1} & D_3^m \\ \hat{D}_4^{m-1} & D_4^m \end{bmatrix} = \begin{bmatrix} \tilde{D}_1^{m-1} & \tilde{D}_1^m \\ \tilde{D}_2^{m-1} & \tilde{D}_2^m \\ \tilde{D}_3^{m-1} & \tilde{D}_3^m \\ \tilde{D}_4^{m-1} & \tilde{D}_4^m \end{bmatrix} Q^{m-1}$$

As a result we get the decomposition $\mathcal{J} = \mathcal{J}^m Q$ with

$$\mathcal{J}^m = \begin{pmatrix} \tilde{D}_1^0 & \tilde{D}_1^1 & \cdots & \cdots & \tilde{D}_1^m \\ \tilde{D}_2^0 & \tilde{D}_2^1 & \cdots & \cdots & \tilde{D}_2^m \\ \tilde{D}_3^0 & \tilde{D}_3^1 & \cdots & \cdots & \tilde{D}_3^m \\ \tilde{D}_4^0 & \tilde{D}_4^1 & \cdots & \cdots & \tilde{D}_4^m \\ L^0 & 0 & \cdots & \cdots & 0 \\ T^0 & L^1 & 0 & \cdots & 0 \\ 0 & T^1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & T^{m-2} & L^{m-1} & 0 \end{pmatrix}.$$

With $\Delta \tilde{s} = Q \Delta s$ the first m parts of the transformed increments can be computed recursively

$$\begin{aligned} \Delta \tilde{s}_0 &= -(L^0)^{-1} \tilde{h}^0; \\ \Delta \tilde{s}_j &= (L^j)^{-1} (-\tilde{h}^j - T^{j-1} \Delta \tilde{s}_{j-1}), \quad j = 1, \dots, m-1. \end{aligned}$$

In order to find the last part $Y = \Delta \tilde{s}_m$ we solve the so-called condensed problem

$$\begin{aligned} \min_Y \quad & \|r_1 + \sum_{i=0}^{m-1} \tilde{D}_1^i \Delta \tilde{s}_i + \tilde{D}_1^m Y\|_\nu, \quad \nu = 1 \text{ or } 2 \\ \text{s.t.} \quad & r_2 + \sum_{i=0}^{m-1} \tilde{D}_2^i \Delta \tilde{s}_i + \tilde{D}_2^m Y = 0 \\ & g_0 + \sum_{i=0}^{m-1} \tilde{D}_4^i \Delta \tilde{s}_i + \tilde{D}_4^m Y = 0 \\ & r_3 + \sum_{i=0}^{m-1} \tilde{D}_3^i \Delta \tilde{s}_i + \tilde{D}_3^m Y \geq 0. \end{aligned} \tag{35}$$

The number of variables in the condensed problem is in the general case the number of parameters plus the number of differential and algebraic variables. The number of constraints is equal to the number of equality and inequality constraints in the original parameter estimation problem and the number of the consistency conditions in (3) $m_{r_2} + m_{r_3} + n_z$.

To recover the original increment, a recursive orthogonal transformation is performed:

$$\Delta s = Q^T \Delta \tilde{s}$$

Remark Note that this way of treating the linear problem requires the (computationally expensive) generation of derivatives of the solution of the DAE wrt parameters and *all* differential and algebraic variables, although the algebraic variables are determined by the consistency conditions and often many of the other variables are determined by point conditions, in particular initial conditions.

Reduced approach In order to reduce the number of evaluations of derivatives to the minimum we may exploit point conditions, e.g. known initial and multipoint conditions, see [46, 17].

Assume for simplicity that part of the equality constraints only depend on variables at one multiple shooting point (including in particular the conditions on the algebraic variables). This results in entries in the linear system of the form:

$$A^i \Delta s^i = a^i, \quad i = 0, \dots, m.$$

In the first step of the reduced approach we evaluate the block A^0 and compute an LQ -decomposition

$$A^0 = [L_A^0, 0]Q_A^0,$$

with Q_A^0 orthogonal and L_A^0 lower triangular. Then the solution manifold can be represented as $\Delta s^0 = \Delta s_N^0 + \Delta s_T^0$,

where $\Delta s_N^0 = (Q_A^0)^T \begin{pmatrix} (L_A^0)^{-1}a^0 \\ 0 \end{pmatrix}$ and $\Delta s_T^0 = (Q_A^0)^T \begin{pmatrix} 0 \\ Y_T^0 \end{pmatrix} =: \mathcal{N}Y_T^0$ with Y_T^0 free.

Now we insert this solution into the first matching condition

$$G^0 \Delta s^0 + H^0 \Delta s^1 = \tilde{h}^0 \quad (36)$$

that then can be rewritten as

$$\mathcal{G}^0 Y_T^0 - H^0 \Delta s^1 = \tilde{h}^0 - G^0 \Delta s_N^0, \quad \mathcal{G}^0 := G^0 \mathcal{N}, \quad (37)$$

and we may apply to the matrix $[\mathcal{G}^0, H^0]$ the decomposition procedure described in the previous section, determine the solution manifold and proceed to the next multiple shooting interval.

The advantages of the reduced approach are obvious. To generate the linearized matching conditions in the form (37), only the matrix $\mathcal{G}^0$ of the directional derivatives of the IVP with respect to the columns of $\mathcal{N}$ and one additional directional derivative $G^0 \Delta s_N^0$ have to be computed. The matrix G^0 itself is not needed.

So the effort for the (costly) computation of derivatives of the solution of the DAE is reduced from $n_y + n_z + n_p$ derivatives to the number of columns of $\mathcal{N}$ plus one derivative corresponding to the inhomogeneity, thus resulting in an effort for derivative generation similar to the single shooting case, while still allowing to incorporate initial guesses at all multiple shooting points.

Especially for parameter estimation in large-scale DAE, resulting e.g. from a semidiscretization of PDEs, with only few degrees of freedom in the initial values this technique proved to be very favorable (see e.g. [27]).

3.7.2 Solution of the condensed problem

In general the condensed problem is of the form

$$\begin{aligned} \min_{Y \in \mathbb{R}^{n_Y}} \quad & f_{l_1} = \sum_{i \in M_1} |A_{1i}Y + c_{1i}|^\nu, \quad \nu = 1, 2 \\ \text{s.t.} \quad & A_{2i}Y + c_{2i} = 0, \quad i \in M_2, \quad A_{3i}Y + c_{3i} \geq 0, \quad i \in M_3, \end{aligned} \quad (38)$$

where A_{ki} denotes an n_Y -row-vector. In case of problem (35) with $M_1 = \{1, \dots, m_1\}$, $M_2 = \{1, \dots, m_{r_2} + n_z\}$ and $M_3 = \{1, \dots, m_{r_3}\}$ the number of variables is $n_Y = n_y + n_z + n_p$, namely the number of initial values plus parameters. In case of the reduced approach the number of equality conditions and variables is reduced by the number of variables that are eliminated due to local equality conditions. Anyway, the number of remaining variables is moderate, but the number m_1 of the components of the cost function, which equals the number of measurements in the parameter estimation problem, may be rather large and the problem is typically dense. For the l_2 case the linear constrained least squares problems are solved by using the equality and active inequality constraints to eliminate the according number of variables from the least squares part and apply a SVD to solve the resulting unconstrained least squares problem in the remaining variables. For numerical stabilisation a transformation of variables can be advisable. An active-set-strategy is applied to treat the inequalities (see [16]).

For the l_1 case problem (38) is equivalent to the problem of linear programming with additional m_1 variables and $2 \times m_1$ inequality constraints

$$\begin{aligned} \min_{Y \in \mathbb{R}^{n_Y}, \xi \in \mathbb{R}^{m_1}} \quad & f_{LP}(Y) = \sum_{i \in M_1} \xi_i, \\ \text{s.t.} \quad & \xi_i - A_{1i}Y \geq c_{1i}, \quad i \in M_1, \\ & \xi_i + A_{1i}Y \geq -c_{1i}, \quad \xi_i \geq 0, \quad i \in M_1, \\ & A_{2i}Y + c_{2i} = 0, \quad i \in M_2, \\ & A_{3i}Y + c_{3i} \geq 0, \quad i \in M_3, \end{aligned} \quad (39)$$

and can in principle be solved by the simplex method of linear programming. However, it is more effective to solve the problem by the dual simplex method. Indeed, the dual problem for (39) and hence for (38) is an “ordinary” bounded-variable problem of linear programming

$$\begin{aligned} \min_{\lambda_k \in \mathbb{R}^{M_k}, k=1,2,3} \quad & \varphi(\lambda) = \sum_{k=1,2,3} c_k^T \lambda_k, \\ \text{s.t.} \quad & \sum_{k=1,2,3} \sum_{i \in M_k} \lambda_{ki} A_{ki}^T = 0, \\ & |\lambda_{1i}| \leq 1, \quad i \in M_1, \quad \lambda_{3i} \geq 0, \quad i \in M_3. \end{aligned} \quad (40)$$

This kind of problems can be very effectively solved by the “long-step” dual simplex method [38], which takes into account the special structure of the inequality constraints in the linear programming problem (39). Details about this method can be found in [39].

It may happen that the matrix A in (38) does not have full rank and then problem (38) may have an unbounded solution. This situation can be avoided by solving a regularized problem

$$\begin{aligned} \min \quad & f(Y) = \sum_{i \in M_1} |A_{1i}Y + c_{1i}|, \\ \text{s.t.} \quad & A_{2i}Y + c_{2i} = 0, \quad i \in M_2, \quad A_{3i}Y + c_{3i} \geq 0, \quad i \in M_3, \\ & L \leq Y \leq U, \end{aligned} \quad (41)$$

which includes lower L and upper U bounds on Y . For the solution of the problem (41) a modification of the “adaptive” method [29] can be effectively applied.

3.8 Inexact l_1 Gauss-Newton Method

In the iterations of a generalized Gauss-Newton method ((9), (10)) we need eventually highly accurate Jacobians, which can be computationally very expensive. In this section we present an inexact l_1 Gauss-Newton method, which does not need exact Jacobians. Instead, in each iteration of this method we need only to evaluate residuals and the gradient of the Lagrange function that can be effectively performed, e.g. by the reverse mode of Automatic Differentiation.

In each iteration of the inexact Gauss-Newton method (9) the increment ΔX^k solves the following problem at $X = X^k$, $\lambda = \lambda^k$

$$\begin{aligned} \min_{\Delta X \in \mathbb{R}^n} \quad & \sum_{i \in I_1} |F_{1i}(X) + \mathcal{A}_{1i} \Delta X| - \sum_{s=1,2,3} \sum_{i \in I_s} \lambda_s^T (\mathcal{A}_{si} - J_{si}(X)) \Delta X, \quad (42) \\ \text{s.t.} \quad & F_{2i}(X) + \mathcal{A}_{2i} \Delta X = 0, i \in I_2, \quad F_{3i}(X) + \mathcal{A}_{3i} \Delta X \geq 0, i \in I_3. \end{aligned}$$

Here the matrix $\mathcal{A} = [\mathcal{A}_{si}]_{i \in I_s, s=1,2,3}$ is an approximation of the Jacobian $\mathcal{J}(X)$; λ^k is a Lagrange vector from the previous iteration.

The following result shows that the inexact Gauss-Newton method ((9), (42)) eventually identifies the correct optimal “active set” of the nonlinear l_1 problem (8).

Theorem 3.13 *Suppose that X^* is a solution of the problem (8) that satisfies the constraint qualification (11) and $n = m_0$, which means that exactly n components of the cost function and constraints in the nonlinear l_1 problem are zero. Assume further that strict complementarity holds. Then if (X^k, λ^k) is sufficiently close to (X^*, λ^*) , the linear l_1 -problem (42) has a unique solution $(\Delta X^k, \lambda_{LP}^k)$ for any fixed matrix $\mathcal{A}$ with*

$$\text{rank } \mathcal{A}_0 = |I(X^*)|, \mathcal{A}_0 = [\mathcal{A}_{si}]_{i \in I_s(X^*), s=1,2,3}. \quad (43)$$

Moreover, the active set $I(\Delta X^k)$ in the problem (42) is the same as the active set $I(X^*)$ of the nonlinear problem (8) at X^* .

Since the active set does not change near a solution satisfying the regularity assumptions, the full-step ($t^k \equiv 1$) inexact Gauss-Newton method ((9), (42)) method becomes equivalent to a Newton-type method applied for solving a system of nonlinear equations (17). This allows us to establish conditions for local convergence of inexact Gauss-Newton method ((9), (42)), see [16].

Theorem 3.14 *Let D be the domain defined in Theorem 3.13 and the matrix $\mathcal{A}$ satisfying (43) is given. Assume further that*

$$\|\mathcal{A}_0^{-1}(\mathcal{F}_0(X) + \mathcal{J}_0(X)\Delta X)\| \leq \kappa \|\Delta X\|, \quad \kappa < 1,$$

for all $X, X + \Delta X \in D$ with $\Delta X = -\mathcal{A}_0(X)\mathcal{F}_0(X) \neq 0$. Then the following holds:

- if $D^0 := B(X^0, \|\Delta X^0\|/(1-\kappa)) \subset D$, then the sequence of iterates defined by ((9), (42)) with $t^k = 1$ starting at X^0 remains in D^0 ,
- there exists $X^* \in D^0$ with $\mathcal{A}_0^{-1}\mathcal{F}_0(X^*) = 0$ and $X^k \rightarrow X^*$ ($k \rightarrow \infty$), and the convergence is linear with

$$\|\Delta X^{k+1}\| \leq \kappa \|\Delta X^k\|.$$

Global convergence of the inexact Gauss-Newton method ((9), (42)) can be ensured by means of classical line search methods under the assumption that the matrix A is a sufficiently good approximation of the Jacobian.

4 Statistical Assessment of the Solution

4.1 Covariance matrix

It is important for parameter estimation problems to compute not only the values of the parameters but also a statistical assessment of the results. This can be done by means of the covariance matrix. A representation of the covariance matrix for *unconstrained* nonlinear parameter estimation problems is well-known, see e.g. [12]. In the following, this notion is generalized to *constrained* parameter estimation problems, see [16, 21].

In this section we consider the nonlinear constrained l_2 parameter estimation problem in the form

$$\min_{X \in \mathbb{R}^n} \quad \frac{1}{2} \sum_{i \in I_1} F_{1i}^2(X), \quad \text{s.t. } F_{Ci}(X) = 0, \quad i \in I_C. \quad (44)$$

Here, as before $I_1 = \{1, \dots, m_1\}$, $I_C = \{1, \dots, m_C\}$, the equality constraints $F_{Ci}(X) = 0$, $i \in I_C$, include a dynamic model, e.g. a discretized boundary value problem in DAEs, $F_{1i}(X)$, $i \in I_1$ represent the weighted difference between model responses and measurements, i.e. the $1/\sigma_{1i}$ weighted measurement errors.

In each iteration of the generalized Gauss-Newton method (9) applied to (44), we solve the linearized problem (23) to compute the increment ΔX^k . We assume throughout this section that the Jacobians $\mathcal{J}_1(X)$ and $\mathcal{J}_C(X)$ satisfy the regularity assumptions (22) on domain D .

Let $\mathcal{J}$ be the Jacobian at the solution X^* and $\mathcal{J}^+$ be the corresponding generalized inverse defined according to (25). Due to the statistical errors of the data as input of the parameter estimation problem, the estimate as the result of the solution procedure is also a random variable. If the measurement errors are normally distributed with zero mean and variances σ_{1i}^2 , then up to first order the estimated solution is normally distributed with expected value X^{true} and variances

$$\mathcal{E}\left((X^* - X^{true})(X^* - X^{true})^T\right). \quad (45)$$

We may approximate a variance-covariance matrix by the following matrix which we will call in the following, for the sake of brevity, a *covariance matrix*

$$C = \mathcal{J}^+ \begin{pmatrix} \mathcal{I}_{m_1 \times m_1} & 0 \\ 0 & 0_{m_C \times m_C} \end{pmatrix} \mathcal{J}^{+T}. \quad (46)$$

Obviously, the matrix C is positive semi-definite matrix with $\text{rank } C = \bar{m} := n - (m_C)$.

Now let us show, how to compute confidence regions for all variables in constrained parameter estimation problems. Generalizing the unconstrained case, we define a nonlinear confidence region for the solution X^* of the nonlinear *constrained* parameter estimation problem (44) by

$$G_N(\alpha) := \{X \mid \mathcal{F}_C(X) = 0, \|\mathcal{F}_1(X)\|_2^2 - \|\mathcal{F}_1(X^*)\|_2^2 \leq \gamma^2(\alpha)\}$$

where $\gamma^2(\alpha) := \chi_{\bar{m}}^2(1 - \alpha)$ is the quantile of the χ^2 distribution for a value $\alpha \in [0, 1]$ with $\bar{m}$ degrees of freedom. The nonlinear confidence region $G_N(\alpha)$ can be approximated by a linearized confidence region

$$G_L(\alpha) := \{X \mid \mathcal{F}_C(X^*) + \mathcal{J}_C(X^*)(X - X^*) = 0, \\ \|\mathcal{F}_1(X^*) + \mathcal{J}_1(X^*)(X - X^*)\|_2^2 - \|\mathcal{F}_1(X^*)\|_2^2 \leq \gamma^2(\alpha)\}.$$

The following lemma gives another, more illustrative, representation of the linear confidence region.

Lemma 4.1 *Let X^* be a solution of problem (44) satisfying the regularity assumptions (22). Then*

$$G_L(\alpha) = \bar{G}_L(\alpha) := \{X^* + \Delta X \mid \Delta X = -\mathcal{J}^+(X^*) \begin{pmatrix} \eta \\ 0 \end{pmatrix}, \|\eta\|_2^2 \leq \gamma^2(\alpha)\}. \quad (47)$$

The next result shows that the linearized confidence region $G_L(\alpha)$ is contained *exactly* in a minimal box, which is the cross product of so-called confidence intervals.

Lemma 4.2 *Let X^* be a solution of parameter estimation problem (44) satisfying the regularity assumptions (22). Then*

$$G_L(\alpha) \subset \bigtimes_{i=1}^n [X_i^* - \theta_i, X_i^* + \theta_i],$$

where $\theta_i = C_{ii}^{\frac{1}{2}} \gamma(\alpha)$. Here C_{ii} denotes the diagonal elements of the covariance matrix C . Furthermore, the following exact bounds hold

$$\max_{X \in G_L(\alpha)} |X_i - X_i^*| = \theta_i, \quad i = 1, \dots, n.$$

Lemma 4.2 shows that the diagonal elements of the covariance matrix play an important role in the statistical assessment of the estimates, namely they are the basis for joint confidence intervals. The proofs can be found in [16, 21].

4.1.1 Covariance matrix as a solution of a linear system

In this section we will show that the covariance matrix $C \in \mathbb{R}^{n \times n}$ for the constrained parameter estimation problem

$$C = \mathcal{J}^+ \begin{bmatrix} \mathcal{I} & 0 \\ 0 & 0 \end{bmatrix} (\mathcal{J}^+)^T \quad (48)$$

satisfies a linear system of equations. For simplicity of notation we omit the dependence of C on the linearization point X in this section. Throughout the section we assume that the matrices $\mathcal{J}_1$ and $\mathcal{J}_C$ satisfy the regularity assumption (22). Let us denote

$$\begin{pmatrix} \mathcal{J}_1^T \mathcal{J}_1 & \mathcal{J}_C^T \\ \mathcal{J}_C & 0 \end{pmatrix}^{-1} := \begin{pmatrix} \mathcal{X} & \mathcal{Y} \\ \mathcal{Z} & \mathcal{T} \end{pmatrix} := \mathcal{M}^{-1} \quad (49)$$

$$\mathcal{X} \in \mathbb{R}^{n \times n}, \mathcal{Y} \in \mathbb{R}^{n \times m_C}, \mathcal{Z} = \mathcal{Y}^T \in \mathbb{R}^{m_C \times n}, \mathcal{T} \in \mathbb{R}^{m_C \times m_C}.$$

Lemma 4.3 *The covariance matrix C from (48) is equal to the matrix $\mathcal{X}$ and thus satisfies the following linear equation system with respect to the variables $C \in \mathbb{R}^{n \times n}$ and $\mathcal{Z} \in \mathbb{R}^{m_C \times n}$*

$$\begin{aligned} \mathcal{J}_1^T \mathcal{J}_1 C + \mathcal{J}_C^T \mathcal{Z} &= \mathcal{I}, \\ \mathcal{J}_C C &= 0. \end{aligned} \quad (50)$$

Proof. According (25) and (48) we have

$$\begin{aligned} C &= \begin{pmatrix} \mathcal{I} & 0 \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \mathcal{J}_1^T \mathcal{J}_1 & 0 \\ 0 & 0 \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \mathcal{I} \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \mathcal{X} & \mathcal{Y} \end{pmatrix} \begin{pmatrix} \mathcal{J}_1^T \mathcal{J}_1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \mathcal{X} \\ \mathcal{Z} \end{pmatrix} = \mathcal{X} \mathcal{J}_1^T \mathcal{J}_1 \mathcal{X}. \end{aligned} \quad (51)$$

Since the blocks $\mathcal{X}$ and $\mathcal{Z}$ of the matrix $\mathcal{M}^{-1}$ satisfy the linear system

$$\begin{aligned} \mathcal{J}_1^T \mathcal{J}_1 \mathcal{X} + \mathcal{J}_C^T \mathcal{Z} &= \mathcal{I}, \\ \mathcal{J}_C \mathcal{X} &= 0, \end{aligned} \quad (52)$$

relation (51) and the symmetry of $\mathcal{X}$ yields

$$C = \mathcal{X}(\mathcal{I} - \mathcal{J}_C^T \mathcal{Z}) = \mathcal{X}$$

which means that $C = \mathcal{X}$. $\diamond$

Remark Note that according to Lemma 4.3 the covariance matrix C is a generalized inverse of the matrix $\mathcal{J}_1^T \mathcal{J}_1$, that is it satisfies $C(\mathcal{J}_1^T \mathcal{J}_1)C = C$. The covariance matrix can be computed by direct linear algebra methods using the decompositions of $\mathcal{J}_1$ and $\mathcal{J}_C$ computed anyway while solving the constrained least squares problem.

It follows from Lemma 4.3 that the columns of the covariance matrix solve special quadratic problems. This fact is used for derivation of a numerical procedure to compute the covariance matrix and in particular its trace based on a conjugate gradient method for linearized parameter estimation problem (23), for details see [21]. Furthermore also derivatives of the covariance matrix C as function of the matrices $\mathcal{J}_1$ and $\mathcal{J}_C$ can be derived. They are needed for efficient numerical methods for optimal experiment design.

5 Experimental design for nonlinear dynamic models

When applying the the statistical analysis in practice, it turns out that data evaluated under different experimental conditions typically lead to estimates of parameters with quite different confidence regions depending on the experimental conditions. Aim is to find those experimental set-ups and operations that result in the best statistical quality for the estimated parameters and at the same time satisfy additional constraints, e.g. limitations of experimental costs, safety requirements, feasibility of experiments, validity of the model etc. Thus the aim of optimum experimental design is to construct a set of N_{ex} experiments by choosing appropriate experimental variables $q = (q_1, \dots, q_{N_{ex}})$, and experimental controls $u = (u_1, \dots, u_{N_{ex}})$ in order to maximize the statistical quality of the variables to be estimated. For this purpose we solve a special optimization problem. Since the covariance matrix C describes an approximation of the confidence region, its "size" may be taken as a cost functional. Typical choices are

$\Phi(C) = \text{trace}(C)$, or $\Phi(C) = \max_i C_{ii}^{\frac{1}{2}}$, which is (up to a factor) the maximal edge of the enclosing box.

Decision variables of the optimization problem are the control profiles $u_i(t)$ (e.g. temperature profiles of cooling/heating or inflow profiles in reaction kinetics experiments, or force and torque inputs in mechanical multibody systems) and the time-independent control variables q_i (e.g. initial concentrations or reactor volume in reaction kinetics, or properties of the experimental device) including integer variables describing whether a possible data evaluation should be carried out or not.

The optimum experimental design approach leads to a control and state constrained optimal control problem, which can be stated in a fairly general form as follows

$$\begin{aligned} & \min_{q,u} \quad \Phi(C(x,p,q,u)), & (53) \\ \text{where} \quad & C \text{ is the covariance matrix of the underlying} \\ & \text{multiple experiment parameter estimation problem,} \\ & x = (x_1, \dots, x_{N_{ex}}), q = (q_1, \dots, q_{N_{ex}}), u = (u_1, \dots, u_{N_{ex}}) \\ \text{s.t.} \quad & c_i(t, x_i(t), p, q_i, u_i(t)) \geq 0, \quad t \in T_i = [t_0^i, t_f^i], \quad i = 1, \dots, N_{ex}, \\ & x_i(t) := (y_i(t), z_i(t)), q_i, u_i(t), p, \quad t \in T_i, \quad \text{satisfy the model dynamics, e.g.} \\ & \dot{y}_i(t) = f_i(t, y_i(t), z_i(t), p, q_i, u_i(t)), \quad i = 1, \dots, N_{ex}, \\ & 0 = g_i(t, y_i(t), z_i(t), p), \quad t \in T_i, \\ & \text{and constraints of the underlying parameter estimation problem} \\ & r_{2,i}(y_i(\theta_1^i), \dots, y_i(\theta_{\mathcal{K}_i}^i), p, q_i, u_i) = 0, \\ & \theta_s^i \in T_i, s = 1, \dots, \mathcal{K}_i, \quad i = 1, \dots, N_{ex}. \end{aligned}$$

The main difficulty of this nonlinear optimal control problem lies in the non-standard objective function that is nonseparable and implicitly defined on the sensitivities of the underlying parameter estimation problem, i.e. on the derivatives of the solution of the differential equation system with respect to the parameters and initial values.

While experimental design for linear models is well established and discussed, e.g. in [3, 28, 43], numerical methods for design of experiments for nonlinear dynamic systems were only developed in the last years, cf. [11, 9, 36, 37].

These numerical methods are based on the direct approach, according to which the control functions are parameterized and the control and state constraints are discretized on appropriate grids. The solution of the DAE systems and its derivatives are assumed to be given as function of initial values and controls. As a result, we obtain a finite-dimensional constrained nonlinear optimization problem, which can be formally written as a general nonlinear programming problem

$$\min_{\xi \in \mathbb{R}^{n_\xi}} \varphi(\xi, s, p) \quad \text{s.t.} \quad \psi_i(\xi, s, p) = 0, i \in \mathcal{E}, \quad \psi_i(\xi, s, p) \leq 0, i \in \mathcal{I}. \quad (54)$$

Here we summarize all experimental design decision variables, including the (parameterized) controls, in the n_ξ -vector ξ . The vector $s \in \mathbb{R}^{n_s}$ denotes a parameterization of the state variables of the process models, p is the (given) n_p -vector of parameters. The functions $\varphi : \mathbb{R}^{n_\xi + n_s + n_p} \rightarrow \mathbb{R}$, $\psi_i : \mathbb{R}^{n_\xi + n_s + n_p} \rightarrow \mathbb{R}$, $i \in \mathcal{E} \cup \mathcal{I}$, are twice-continuously differentiable. Note that the equality constraints $\psi_i(\xi, s, p) = 0, i \in \mathcal{E}$, contain the discretized boundary

value problem. Problem (54) is solved by a structure-exploiting SQP method, where the integer variables are treated in a relaxed form. The main effort for the solution of the optimization problem by the SQP method is spent on the calculation of the values of the objective function and the constraints as well as its derivatives. Note that second order derivatives are required, since the objective function is based on the first derivatives already. Efficient methods for the solution of the DAE systems and for the generation of its first and second derivatives have been developed, e.g. in [9], combining Internal Numerical Differentiation [16] of the DAE solution and Automatic Differentiation [33] of the model functions.

For a more detailed discussion of the numerical methods for nonlinear optimum experimental design see [9, 10, 11, 36] and [37], where also methods for the design of robust optimal experiments are described. They allow to compute worst-case designs for the case that the parameter values are only known to lie in a possible large confidence region.

The treatment of many applications in particular in chemical reactions [11], enzyme reactions [20] and soil physics [2] show that drastic improvements of the quality of parameter estimates can be achieved.

6 Numerical Results

Notorious test example

The parameter estimation problem considered here is a modification of a two-point boundary problem from [25]. The dynamics is described by an ODE system with two states and one parameter p

$$\dot{x}_1 = x_2 \quad \dot{x}_2 = \mu^2 x_1 - (\mu^2 + p^2) \sin pt \quad t \in [0, 1]$$

and initial values

$$x_1(0) = 0 \quad x_2(0) = \pi.$$

The solution for the true parameter value $p = \pi$ is

$$x_1(t) = \sin \pi t, \quad x_2(t) = \pi \cos \pi t.$$

Measurement data is simulated using this solution $x(t)$ at the measurement points $t_j = 0.1 \times j$ $j = 1, \dots, 10$ and pseudo-random normally distributed errors.

This test problem is a remarkable example for the superiority of the BVP approach vs the IVP approach. Indeed, even using the true value of the parameter (up to 16 digits) we cannot perform the numerical integration throughout the whole integration interval, see Figure 2 showing simulations for $p = 1$ and $p = \pi$ for $\mu = 60$.

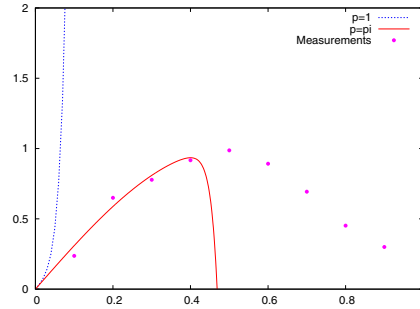


Fig. 2 IVP approach: Initial trajectories for $p = 0$ and $p = \pi$

The problem is not numerically solvable with the IVP approach. This is not due to a failure of the integration procedure. The eigenvalues of the stability matrix of the system are $\lambda_{1,2} = \pm\mu$, and the general solution is given by

$$\begin{aligned} x_1(t) &= \sin pt + \varepsilon_1 \sinh \mu t + \varepsilon_2 \cosh \mu t; & \varepsilon_1 &= \frac{x_2(0) - p}{\mu} \\ x_2(t) &= p \cos pt + \varepsilon_1 \mu \cosh \mu t + \varepsilon_2 \mu \sinh \mu t; & \varepsilon_2 &= x_1(0). \end{aligned}$$

Thus, even small disturbances in the initial values for the states and the parameter and unavoidable discretization and rounding errors are drastically amplified and eventually dominate the solution.

Applying the BVP approach, the problem is solved in only 4 iterations for rather high values of μ . Fig. 3 shows initial and solution trajectories for $\mu = 60$.

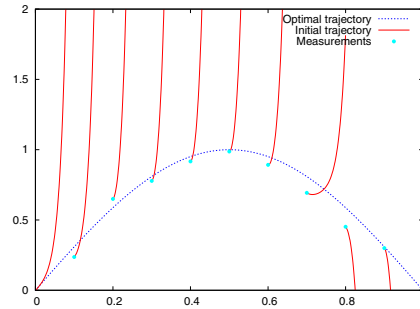


Fig. 3 BVP approach: Initial and solution trajectories

The Light Reaction in Photosynthesis

In this section we consider the fast transient of chlorophyll fluorescence, which is used as an important step in photosynthesis and an object of intensive research, see e.g. [6]. It results from the interplay of two aspects of the light reaction: the electron transport chain and the distribution of excitation energy within the thylakoid membrane. In the following we consider

a version of a model developed by Baake in her thesis [4], who also performed the experiments and evaluated the considered data. The process is described by the following system of six differential equations:

$$\begin{aligned}
 \dot{y}_1 &= (k_a + k_3(p - y_6))y_1 + k_{-3}y_5y_6 \\
 \dot{y}_2 &= k_a y_1 - (k_1 + k_4(p - y_6))y_2 + k_{-1}y_3 + k_{-4}y_6(1 - \sum_{i=1}^5 y_i) \\
 \dot{y}_3 &= k_1 y_2 - (k_a + k_{-1})y_3 \\
 \dot{y}_4 &= k_a y_3 - k_2 y_4 + k_{-2}y_5 \\
 \dot{y}_5 &= k_3 y_1(p - y_6) + k_2 y_4 - (k_a + k_{-2} + k_{-3}y_6)y_5 \\
 \dot{y}_6 &= -k_{-3}(1 - \sum_{i=1}^5 y_i)y_6 + k_3(y_1 + y_2)(p - y_6) + (p - y_6)k_{\text{lim}}
 \end{aligned}$$

with initial values

$$\begin{aligned}
 y_1(0) &= 3/4, y_2(0) = 0.25k_{-1}/k_1, y_3(0) = 0.25(1 - k_{-1}/k_1) \\
 y_4(0) &= y_5(0) = 0, y_6 = p.
 \end{aligned}$$

Here the light reaction rate is given by

$$k_a = \frac{J_2(1 - p_{2T})}{1 - p_{22} - p_{2T} + p_{22}p_{2T}(y_1 + y_3 + y_5)}.$$

The measurement function describes the observation of fluorescence:

$$F(t) = \left\{ \frac{1 - p_{2T} - p_{22}}{p_{2T}} + \frac{1 - (y_1 + y_3 + y_5)}{1 + \frac{p_{22}p_{2T}(y_1 + y_3 + y_5)}{1 - p_{2T} - p_{22}}} \right\} \cdot S \cdot J_2$$

For the quantification of the model the fluorescence rise of dark-adapted living tobacco leaves after exposure to light is measured.

It is assumed that $k_3 = k_{-3} = k_4 = k_{-4}$ and that the values of k_1 , k_{-1} , k_2 and k_{-2} are fixed, which leaves us with the following eleven parameters, namely four system parameters J_2 , k_3 , k_{lim} , p , one observation parameter S and two experiment dependent parameters p_{22} , p_{2T} for each experiment.

Data of three experiments with 96 measurements each, conducted at different light intensities are used in the multiple experiment parameter estimation problem.

With the following initial values for the parameters

	k_3	p_{22}	p_{2T}	p	S	J_2	k_{lim}
Exp 1	20	0.5	0.4	10	10	300	1
Exp 2						210	1
Exp 3						150	1

and 20 multiple shooting points, the following 3 experiment multiple shooting trajectories for the measured quantity result (Fig. 4). Note that we use initial guesses at the multiple shooting nodes that exactly reproduce the measurements.

As result of the parameter estimation, after 12 iterations we get the following values of the parameters and the corresponding standard errors:

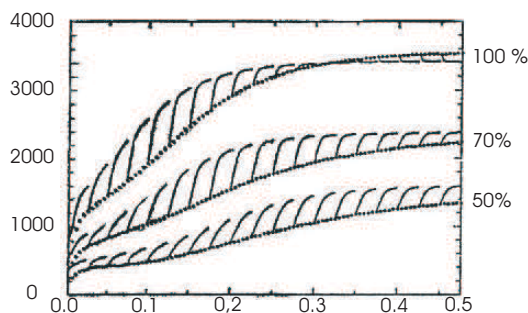


Fig. 4 Photosynthesis: Initial trajectories for multiple shooting with 20 gridpoints; vertical axis: fluorescence; horizontal axis: time in sec; dots represent the measurements

	k_3	p_{22}	p_{2T}	p	S
solution	17.3	0.0710	0.841	11.5	18.1
standard error ¹	± 0.76	± 0.015	± 0.015	± 0.84	± 1.0

	100%		70%		50%	
	J_2	k_{lim}	J_2	k_{lim}	J_2	k_{lim}
solution	195.	1.07	143.	1.92	101.	1.67
standard error ¹	± 9.0	± 0.36	± 6.9	± 0.26	± 5.3	± 0.17

¹ estimated; multiplication by 4.5 yields 95% confidence intervals

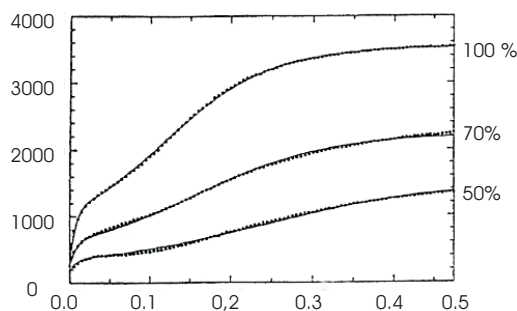


Fig. 5 Photosynthesis: 3-experiment solution

6.1 Biochemical Problem

Here we consider a biochemical problem from enzyme kinetics. The model involves four chemical species, one feed stream and a temperature control. The kinetics is described by Arrhenius mass action kinetics. Eight unknown parameters have to be estimated from only one indirect measurement function. Details about the model can be found in [24] and [20].

This real-life problem turned out to be very difficult, since information from one experiment is insufficient for the identification of all parameters, and since the measurement error

is large. To improve the information content of the data we first have applied the new nonlinear optimum design methods described in chapter 5 to design 5 supplementary temperature profiles to achieve identifiability.

Using the designed temperature profiles the measurements have been simulated without outliers (A), with outliers (B), with large outliers (C). Table 1 presents the “true” and the estimated values of the parameters using l_2 based and l_1 based parameter estimation respectively. The number of iterations necessary for solving the estimation problems are given in Table 2. We may conclude that the l_1 based parameter estimation shows better performance than the traditional l_2 approach.

Table 1 Biochemical reaction: true and estimated parameter values

true values	20.31	32.19	0.591	208.9	-2.52	-24.06	-4.54	-12.91
A / l_2	20.29	32.14	0.587	207.5	-2.53	-23.89	-4.56	-12.88
A / l_1	20.28	32.13	0.585	206.9	-2.52	-23.85	-4.56	-12.88
B / l_2	22.08	36.92	0.588	207.3	-2.19	-28.51	-4.67	-11.52
B / l_1	20.17	31.82	0.587	207.8	-2.58	-22.81	-4.60	-12.73
C / l_2	20.11	31.69	1.149	408.3	-4.95	4.63	-6.07	-10.67
C / l_1	20.12	31.67	0.528	186.9	-2.51	-22.97	-4.75	-12.67

Table 2 Biochemical reaction: l_2 vs l_1 , number of iterations

	l_2	l_1
A	2	2
B	6	4
C	7	3

7 Summary

We have presented an overview on numerical methods for parameter estimation in differential algebraic equation systems. Emphasis is put on the numerically stable boundary value problem approach, recent developments in robust l_1 estimation methods and a statistical assessment of the estimated results.

Newly developed methods for the design of optimal experiments for parameter estimation that allow to significantly improve the statistical quality of the parameter estimates and/or reduce the experimental costs are sketched.

Results on parameter estimation problems demonstrate some of the features of the methods.

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